DATA SCIENCE AND VISUALIZATION

Project Planning and Dataset Selection

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team Number: Team 9

1. INTRODUCTION

The first week focused on team formation, dataset selection, defining the problem statement and objectives, and planning the project workflow. The team selected a Spotify tracks dataset containing information about songs, artists, albums, popularity, and audio features.

The team also learned about Pandas and NumPy for data analysis and identified the tools required for the project. These activities established the foundation for exploring the relationship between musical characteristics and song popularity in the upcoming stages.

2. PROJECT TITLE

Discovering the Factors Behind Song Popularity Through Data Science

3. PROBLEM STATEMENT

Music streaming platforms contain a large number of songs with different musical characteristics and popularity levels. Understanding the characteristics associated with song popularity can help reveal meaningful patterns in music data.

This project aims to analyze Spotify track data using data science and visualization techniques. The project will involve understanding the dataset, assessing its quality, handling missing values and inconsistencies, performing exploratory data analysis, and investigating relationships between audio features and popularity.

The findings will be communicated through suitable visualizations and, in the later stages, an interactive dashboard.

4. OBJECTIVES

The main objectives of the project are:

- To select and understand the Spotify dataset and its attributes.
- To learn Pandas and NumPy for data analysis.
- To identify data-quality issues and perform exploratory data analysis.
- To analyze relationships between musical features and popularity, create visualizations, and prepare data for advanced analysis and dashboard development.

5. TEAM FORMATION

The project team consists of four members who will collaboratively work on dataset preparation, analysis, visualization, documentation, and presentation.

S. No.	Team Member	Register Number
1	Prithesa R	25BAD083
2	Saravanan A	25BAD102
3	Tejas T	25BAD124
4	Viknesh T	25BAD128

The team will share responsibilities to ensure that each stage of the project is completed systematically. Specific individual task assignments can be documented once finalized by the team.

6. DATASET SELECTION

6.1 Dataset Description

The team selected the Spotify tracks dataset, containing 19,335 records and 25 columns covering song, artist, album, popularity, and audio characteristics.

6.2 Important Attributes

The dataset contains the following categories of information:

- **Track information:** Track ID, track name, track number, and album details.
- **Artist information:** Artist names associated with tracks.
- **Album information:** Album name, album type, release date, and number of tracks in an album.

- **Popularity information:** Popularity values associated with tracks.
- **Audio features:** Danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo.
- **Musical attributes:** Key, mode, and time signature.

These attributes provide a combination of numerical, categorical, and textual data that can be examined using statistical and visualization techniques.

6.3 Reasons for Dataset Selection

The Spotify dataset was selected for the following reasons:

1. Represents real-world music data with details about songs, artists, and popularity.
2. Contains multiple attributes and audio features suitable for statistical and comparative analysis.
3. Provides opportunities to practise data cleaning, preprocessing, exploratory data analysis, correlation analysis, and visualization.
4. Supports advanced techniques such as feature engineering and Principal Component Analysis (PCA).

7. INTRODUCTION TO PANDAS

7.1 Definition

Pandas is an open-source Python library used for data manipulation and analysis. It provides two important data structures: Series and DataFrame.

A Series represents one-dimensional labelled data, whereas a DataFrame represents two-dimensional tabular data arranged in rows and columns.

7.2 Applications of Pandas

Pandas provides several useful operations for working with datasets:

1. Loading datasets from CSV and other file formats and displaying the first and last few records.
2. Examining the number of rows and columns, column names, and data types.
3. Identifying missing values and duplicate records and generating descriptive statistics.
4. Filtering, sorting, and transforming data for further analysis.

7.3 Application in Our Project

Pandas will be used to load and inspect the Spotify dataset. It will help the team understand the dataset structure, identify data-quality issues, and prepare the data for cleaning and exploratory analysis.

Example:

```
import pandas as pd
df = pd.read_csv("DSV_TEAM-9_RAW_DATASET.csv")
print(df.head())
print(df.shape)
```

The read_csv() function loads the dataset, head() displays the first five rows by default, shape returns the number of rows and columns.

8. INTRODUCTION TO NUMPY

8.1 Definition

NumPy, which stands for Numerical Python, is a Python library used for numerical computing. It provides the multidimensional array structure and mathematical functions required for efficient numerical operations.

8.2 Applications of NumPy

NumPy supports:

- Creating and manipulating numerical arrays for data processing.
- Performing mathematical and statistical calculations, including mean, median, and standard deviation.
- Carrying out element-wise operations and numerical transformations.
- Supporting data preprocessing, statistical analysis, and future PCA workflows.

8.3 Application in Our Project

NumPy will be used to perform numerical calculations on relevant Spotify audio features. It can help calculate statistical measures and prepare numerical data for further analysis.

Example:

```
import numpy as np
audio_features = np.array([0.75, 0.82, 0.68, 0.90, 0.73])
```

```
print("Average Audio Feature:", np.mean(audio_features))
print("Standard Deviation:", np.std(audio_features))
```

The `array()` function creates a numerical array, `mean()` calculates its average, and `std()` calculates its standard deviation.

9. PROPOSED PROJECT METHODOLOGY

The project will follow a systematic data science workflow to study Spotify song popularity and explore relationships between musical characteristics and popularity.

The planned workflow is:

1. **Raw Dataset Selection:** Select the Spotify tracks dataset for the project.
2. **Data Understanding:** Study the dataset structure, attributes, and meaning of the available information.
3. **Data Cleaning:** Plan the preparation of the dataset for reliable analysis.
4. **Exploratory Data Analysis:** Explore patterns and relationships among relevant variables.
5. **Feature Engineering:** Prepare and transform suitable attributes for analysis.
6. **Feature Selection:** Identify relevant features for investigating song popularity.
7. **Principal Component Analysis (PCA):** Explore dimensionality reduction where appropriate.
8. **Data Visualization:** Develop charts and visual representations of the findings.
9. **Insights and Data Storytelling:** Present the major findings in an understandable form.

12. TOOLS AND TECHNOLOGIES

The following tools and technologies are planned for the project:

- **Python:** Primary programming language for data analysis.
- **Pandas:** Data manipulation and handling tabular datasets.
- **NumPy:** Numerical calculations and mathematical operations.
- **Matplotlib:** Creating graphs and charts.
- **Seaborn:** Statistical data visualization.
- **Scikit-learn:** Supporting feature processing and PCA.
- **Jupyter Notebook:** Writing, executing, and documenting Python code.
- **GitHub:** Version control and team collaboration.
- **Streamlit or Plotly:** Potential tools for developing an interactive dashboard.

These tools will support the different stages of the project, from data preparation to visualization and presentation.

13. WEEK 1 ACTIVITIES AND MILESTONES

The main activities planned and undertaken during Week 1 were:

- Formed the project team, selected the Spotify tracks dataset, and finalized the project title.
- Defined the problem statement, project objectives, and overall project scope.
- Introduced Pandas and NumPy and identified the required software tools and libraries.
- Planned the project workflow, established GitHub for collaboration, and prepared the initial documentation.

These activities established the foundation for the subsequent stages of the project.

14. EXPECTED OUTCOMES

The expected outcomes of the project are:

- Understand the Spotify dataset and its attributes.
- Analyze the relationship between audio features and song popularity.
- Create visualizations to identify trends and important features.
- Develop an interactive dashboard to present the findings and insights.

15. CONCLUSION

Week 1 established the foundation for the Data Science and Visualization project titled *Discovering the Factors Behind Song Popularity Through Data Science*.

The team selected the Spotify tracks dataset, defined the problem statement and objectives, and planned the overall project methodology. The team also introduced Pandas and NumPy as essential Python libraries for data manipulation and numerical computation. GitHub was identified as the platform for maintaining project files and supporting collaboration.

These initial activities provide a clear direction for the project. In the following weeks, the team will proceed with the scheduled stages of dataset understanding, preparation, analysis, and visualization to investigate the factors associated with Spotify song popularity.